

Research evolution — review package

Status and provenance

Prepared October 6, 2026 with Codex (GPT-6) assistance. The original repository memo is reproduced unchanged below and separately as `Edition-A-original.md`. Its historical checkpoint field remains unchanged. On October 6 Linda confirmed the original-file match and that the Lab 03 screenshot serves as her checkpoint evidence: Attempt 1, September 1, 4:52-5:42 PM, 25/25. See evidence/edition-a-confirmation.md and edition-a-checkpoint.png. Screenshot timezone and separate receipt identifier are not shown. This package does not attest to pre-AI authorship or reconstruct missing history.

Edition A — preserved text

Project 1 - Edition A

Decision and user

This memo supports the fund's investment committee in deciding whether to initiate a position in Eli Lilly and Company (NYSE: LLY), defer action pending additional work, or decline the opportunity. The committee needs a decision-useful view of the operating thesis, valuation, evidence quality, and what could disprove the case.

Company and valuation date

- Company: Eli Lilly and Company
- Ticker: NYSE: LLY
- Filing reviewed: Form 10-K for the fiscal year ended December 31, 2025
- SEC accession number: 0000059478-26-000013
- Filing date: February 12, 2026
- Valuation date: September 1, 2026
- Reference share price: \$1,160.00
- Reference equity value: approximately \$1.043 trillion, calculated as \$1,160.00 multiplied by 899.3 million diluted weighted-average shares
- Reference enterprise value: approximately \$1.078 trillion, calculated as equity value plus \$42.503 billion of debt less \$7.268 billion of cash

The equity and enterprise values are preliminary reference values, not price targets. The diluted weighted-average share count is an earnings denominator rather than an exact spot-date fully diluted share count, so the valuation must be rebuilt before a final recommendation.

Initial thesis

****Watch-defer.****

Lilly has exceptional operating momentum, led by Mounjaro and Zepbound, and the 2025 results demonstrate substantial revenue and earnings growth. However, the reference valuation already

capitalizes a very large amount of future success. Edition A does not yet establish that risk-adjusted future cash flows justify an enterprise value of roughly \$1.08 trillion. I would place Lilly on the active research list but not initiate a position until the work below establishes an acceptable expected return and downside case.

Known evidence and sources

1. Lilly reported 2025 revenue of \$65.179 billion, up 45% from \$45.043 billion in 2024. Management states that revenue growth was primarily driven by increased volume, partially offset by lower realized prices; Mounjaro and Zepbound were the main drivers of both effects. Source: 2025 Form 10-K, MD&A, page 43; [SEC filing](<https://www.sec.gov/Archives/edgar/data/59478/000005947826000013/lly-20251231.htm>).
2. Net income was \$20.640 billion and diluted EPS was \$22.95 in 2025, versus \$10.590 billion and \$11.71 in 2024. Source: 2025 Form 10-K, Consolidated Statements of Operations, page 57.
3. Weighted-average diluted shares used for 2025 EPS were 899.3 million. Source: 2025 Form 10-K, Consolidated Statements of Operations, page 57.
4. Cash and cash equivalents were \$7.268 billion at December 31, 2025. Source: 2025 Form 10-K, Consolidated Balance Sheets, page 59.
5. Total debt was \$42.503 billion, comprising \$1.635 billion current and \$40.868 billion long-term. Source: 2025 Form 10-K, Debt note, page 76.
6. Net cash provided by operating activities was \$16.8 billion in 2025, up from \$8.8 billion in 2024. Source: 2025 Form 10-K, Financial Condition and Liquidity, page 52.
7. Lilly's long-term success depends on discovering or acquiring, developing, and commercializing innovative medicines. The pipeline listed in the filing includes additional tirzepatide indications as well as late-stage or submitted programs such as orforglipron and insulin efsitora alfa. Source: 2025 Form 10-K, Clinical Development Pipeline, pages 43-47.
8. The filing metadata confirms the entity, reporting period, accession number, and filing date. Source: [SEC filing header](<https://www.sec.gov/Archives/edgar/data/59478/000005947826000013/0000059478-26-000013-index-headers.html>).
9. The September 1, 2026 market reference was \$1,160.00 per share. Source: current LLY market quote captured on the valuation date.

All financial-statement tables present dollars in millions except per-share data; share counts are in millions. Currency is U.S. dollars. The locally preserved source filing is `Lily 10K.pdf` in this project folder.

Consequential assumptions

- Mounjaro and Zepbound volumes can continue growing fast enough to offset lower realized prices and increasing competition.
- Lilly can expand manufacturing capacity without material execution failures, prolonged shortages, or unacceptable capital intensity.
- Tirzepatide's clinical profile and additional indications will protect demand as competing incretin and oral obesity therapies enter the market.
- Payers and governments will permit economics that support durable margins despite pressure to expand access and reduce prices.

- Pipeline productivity beyond tirzepatide will contribute meaningful risk-adjusted value rather than leaving valuation dependent on two brands sharing one molecule.
- The preliminary share count is close enough for triage; a final model will use spot-date basic shares plus all dilutive securities.
- Debt is treated at carrying value and cash as fully available. A final enterprise-value bridge may need leases, noncontrolling interests, investments, pension items, and other claims or non-operating assets.
- A suitable required return and terminal-growth framework will show enough upside from the September 1, 2026 price to compensate for clinical, regulatory, pricing, and concentration risks.

Unknowns

- The revenue, margin, reinvestment, and free-cash-flow growth already implied by the current stock price.
- Current prescription trends, market share, realized net pricing, patient persistence, and supply constraints for Mounjaro and Zepbound.
- Competitor launch timing and comparative efficacy, tolerability, convenience, supply, and price.
- Manufacturing capacity by site and product, ramp timing, bottlenecks, and required capital expenditures.
- Payer coverage, rebates, formulary position, and the effect of policy changes across major markets.
- Probability-adjusted value, development cost, timing, and commercial overlap for key pipeline programs.
- Normalized tax rate, operating margin, working capital, capital intensity, and stock-based compensation needed to convert accounting earnings into owner cash flow.
- Exact spot-date fully diluted shares and a complete enterprise-value bridge.
- Position sizing, liquidity, benchmark constraints, and the committee's required return and maximum acceptable drawdown.

Research plan

1. Build a reverse DCF from the September 1, 2026 enterprise value to quantify the revenue growth, operating margin, reinvestment, and terminal assumptions embedded in the price.
2. Construct bull, base, and bear product models for Mounjaro and Zepbound using patients, dose, persistence, gross-to-net pricing, geographic access, capacity, and competing products.
3. Map manufacturing capacity and announced expansions to demand scenarios; test whether volume growth is physically achievable and what capital intensity it requires.
4. Review the latest 10-Q, earnings call, trial disclosures, prescription data, payer decisions, and competitor primary sources to update the 10-K baseline through the valuation date.
5. Build a probability-adjusted pipeline model for material late-stage assets and additional tirzepatide indications, explicitly separating shared-molecule concentration from independent pipeline value.
6. Reconcile net income to normalized free cash flow and complete the enterprise-value bridge using spot-date dilution and all material financial claims and non-operating assets.

7. Compare expected return and downside across scenarios with the committee's hurdle rate, portfolio exposures, and risk limits.
8. Return with an initiate, continue-to-watch, or do-not-initiate recommendation, a price range, catalysts, disconfirming evidence, and proposed position size.

Partner-generated falsification question

> What evidence would falsify the thesis that Mounjaro and Zepbound can sustain Lilly's current valuation if obesity-market competition, pricing pressure, or manufacturing constraints cause their combined risk-adjusted cash-flow growth to fall materially below the level implied by the stock price?

This question was generated by a partner. The partner did not write the answer, and Edition A intentionally does not answer it.

Checkpoint receipt

Receipt identifier: Pending submission and verification.

Dated addendum — October 6, 2026

Retain September 1 Edition A unchanged. Final model date is September 8 to align the saved peer-price snapshot; statement history remains FY2025 and latest bridge balances June 2026. None of the later analysis is backdated. Prior \$589 FCFE teaching result and \$827 standalone FCFF result use different forecasts/timing and are not interchangeable with this integrated result. The prior Lab 4 \$60m short-term-investments bridge addition is not separately identifiable on the Q2 balance sheet: reject it as an unsupported standalone asset, to avoid potential double counting.

Edition B — conditional committee view

Watch/defer remains the proposed action, supported by the integrated base \$754.19, downside \$398.09, upside \$1228.42. Linda confirmed this judgment and the corrected assumption review on October 6; see Student-Review.md. The preserved initial concern about the growth expectations embedded in price is quantified by the reverse DCF. Read the decision memo for triggers and limitations.

| Item | Edition A | Edition B | Cause and evidence |

|---|---|---|---|

| Action | Watch/defer, no demonstrated DCF support | Watch/defer pending stronger expected-return evidence | Mixed: original thesis, course-required valuation, integrated cash-flow outputs |

| Date/share basis | September 1; 899.3m annual EPS denominator | September 8; 893.7m Q2 diluted proxy for DCF | Primary filing update; spot dilution remains unknown |

| Model | Research plan and preliminary bridge | Six modeled annual statements, 2026 bridge plus five full FCFF years | Course requirement plus AI-assisted implementation; accounting tests |

| Valuation | Reference EV about \$1.08tn, no fair-value target | Base EV \$716.1bn and causal scenarios | Reinvestment, taxes, operating forecast, discount timing |

| Peer challenge | Partner falsification question preserved | October 7 R&D challenge reported by Linda; prior inventory challenge separate | Partner-Discussion.md; verified numerical corrections

and unsourced peer claims disclosed |

| Risks | Competition, pricing, capacity, pipeline | Same risks plus terminal share and fixed-driver feedback limitations | Sensitivities and reverse DCF; no material thesis reversal |

Student assumption review: Linda confirmed the defend/revise/evidence-needed judgments in Student-Review.md. No AI-off prediction or fabricated experiential statement is provided.

October 7 partner discussion addendum

Linda reports an in-person discussion with Annika Rao challenging constant 20% recurring R&D and proposing 17% (16%–18% discussion range). Linda retained 20% and watch/defer. The previously executed 17% exploratory test gives \$810.23/share, not the \$851.07 in supplied notes. No separate October 7 run or cross-driver range sweep was supplied. Peer benchmarks require sourcing; see docs/Partner-Discussion.md.